

Quality Inspection Report QIR-2025-0142

Dimensional / Lot Acceptance Inspection | MANDREL, PACKER, 4-1/2 (Dwg D-PKR-4500-02 Rev C)

Report No. / Date	QIR-2025-0142 / 2025-04-22
Part Number	ACT-PKR-4500-02
Part Description	MANDREL, PACKER, 4-1/2 (Dwg D-PKR-4500-02 Rev C)
Work Order / PO	WO-25-1187 / Stock
Lot / Heat-Batch	L25-044 / 8Q4412
Lot Qty / Sample Qty	12 / 3 (per ACT-QP-08)
Serial Numbers	M45-25-0331 / -0332 / -0333
Material	AISI 4140 Q&T; 28-32 HRC (MTR heat 8Q4412, Lone Star Specialty Metals)
Inspected By	D. Whitfield
Reviewed / Approved By	S. Nguyen
Applicable Specs	API 11D1 / API 19V (as applicable), NACE MR0175/ISO 15156, ASTM E709, API 5B

Dimensional & Attribute Results

Actuals are the worst-case (closest-to-limit) reading across the sample unless noted. Instruments are calibrated per ACT-CAL-01; calibration records on file.

Char.	Feature / Characteristic	Nominal	Tol +	Tol -	Actual (worst)	Instrument	A/R
D1	Body OD	3.780	+0	--0.005	3.785	Micrometer M-212	ACC
D2	Bore ID	1.938	+0.005	-0	1.941	Bore gauge B-108	ACC
D3	Seal bore ID (upper)	2.750	+0.002	-0	2.751	Air gauge A-301	ACC
D4	Seal bore ID (lower)	2.750	+0.002	-0	2.751	Air gauge A-301	ACC
D5	Element seat OD	3.250	+0	--0.003	3.253	Micrometer M-212	ACC
D6	Overall length	41.2	+0.0	--0.0	41.3	Tape/height gauge H-14	ACC
D7	Thread pitch dia, 2-7/8 EUE pin	3.100	+0.004	--0.004	3.104	Thread gauge T-EUE-287	ACC
D8	Surface finish, seal bores (Ra uin)	16.0	+16.0	--8.0	28.4	Profilometer P-52	ACC
D9	Hardness (HRC)	30.0	+2.0	--2.0	32.0	Hardness tester HR-3	ACC
D10	Concentricity OD to bore (TIR)	0.000	+0.006	-0	0.002	V-block + indicator	ACC

Result: ACCEPT

Measuring & Test Equipment Used

Instrument	Cal. Record	Cal. Due	Status
Micrometer M-212	CAL-8385	2026-02-24	In calibration at time of use
Bore gauge B-108	CAL-6641	2026-08-14	In calibration at time of use
Air gauge A-301	CAL-6064	2026-12-02	In calibration at time of use
Tape/height gauge H-14	CAL-8254	2026-07-04	In calibration at time of use

Instrument	Cal. Record	Cal. Due	Status
Thread gauge T-EUE-287	CAL-9902	2026-10-23	In calibration at time of use
Profilometer P-52	CAL-1794	2025-02-06	In calibration at time of use
Hardness tester HR-3	CAL-1121	2026-09-18	In calibration at time of use
V-block + indicator	CAL-0371	2025-01-28	In calibration at time of use

Sampling & Methods

Sampling per ACT-QP-08 (ANSI/ASQ Z1.4 basis). Dimensional measurements at 68-72 degF after minimum 4 hr thermal soak. Surface finish per ASME B46.1. Hardness per ASTM E18 (HRC/HRB) or ASTM E140 conversion where noted. Threads gauged per API 5B with certified working gauges. Magnetic particle inspection, where applied, per ASTM E709 using wet fluorescent method, 100-125 V/in. field.

Inspector Notes

Sample plan per ACT-QP-08 Level II (3 of 12). All characteristics within drawing tolerance. MPI wet fluorescent per ASTM E709: no relevant indications. Thread gauged per API 5B. Material certs on file: heat 8Q4412 (see MTR-LSM-8Q4412).

Sign-off

Role	Name	Signature	Date
QC Inspector (Level II)	D. Whitfield	/signed electronically/	2025-04-22
QA Manager	S. Nguyen	/signed electronically/	2025-04-22